

One Wrong Move: Sudoku + Minesweeper Report

Report 013 · Branch `sudoku-minesweeper-puzzles` · Created 2026-05-03 · Static HTML/CSS/JS app

Summary

Implemented compact Sudoku-inspired and Minesweeper-inspired puzzle families for the mobile-first One Wrong Move app. The new puzzles preserve Ladder Run, Three-Set Free Play, target clarity, the symbol catalog, the puzzle lab, and static deployment. Full 9x9 Sudoku was intentionally avoided; the production Sudoku boards use 4x4 mini-Sudoku constraints that fit the phone shell.

Repo State Summary

- Working folder: `/Users/kevinhegg/Desktop/one-wrong-move`
- Feature branch: `sudoku-minesweeper-puzzles`
- Base before this work: `949d45a Merge symbol library and theme puzzles`
- Nested repo check: only the root `.git` folder was present.
- Main branch status at report creation: pending feature commit, merge, and push after validation.

Files Created Or Changed

- `README.md`
- `docs/game-design.md`
- `docs/pattern-systems.md`
- `docs/symbol-system.md`
- `index.html`
- `public/index.html`
- `public/style.css`
- `public/game.js`
- `public/symbols.js`
- `public/puzzles.js`
- `public/lab.html`
- `public/lab.js`
- `scripts/validate-sudoku-minesweeper.js`
- `docs/codex-reports/screenshots/013-*.png`
- `docs/codex-reports/013-sudoku-minesweeper-report.pdf`

Sudoku Design Rationale

Sudoku appears as mini-Sudoku because the full desktop puzzle is too large for a fast phone-first timed game. The 4x4 version keeps the recognizable rule: every row, column, and 2x2 box must contain digits 1-4 exactly once.

Puzzle	Mode	Target	Validator	Ladder placement
Sudoku Conflict	identifyOne	Tap the wrong digit	Brute-forces every one-cell repair; exactly one cell must restore all rows, columns, and boxes.	Early/mid; difficulty 2.
Mini Sudoku Swap	multiSelect	Select exactly two swapped digits, then Submit Swap	Brute-forces every two-cell swap; exactly one pair must restore the solved mini-Sudoku.	Level 4+; difficulty 3.

Minesweeper Design Rationale

Minesweeper uses a 5x5 board with clue numbers and hidden candidate squares. Clues count mines in the eight surrounding cells. Clue cells are disabled and hidden cells are selectable. Active boards never reveal bombs; flags appear as planned selections or answer reveal states.

Puzzle	Mode	Target	Validator	Ladder placement
Minesweeper Forced Mine	chooseOne	Tap the one hidden square that must be a mine	Enumerates clue-consistent layouts; exactly one hidden square must be mined in every valid layout.	Level 4+; difficulty 3.

Minesweeper Mark All	multiSelect	Select every mine, then Submit Flags	Enumerates layouts and requires exactly one complete mine set.	Later Ladder; difficulty 4.
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Selection Placement

- Ladder avoids retired puzzles and avoids back-to-back number-grid puzzles when alternatives exist.
- Mini Sudoku Swap and Minesweeper Forced Mine are held until Level 4 or later.
- Minesweeper Mark All is reserved for later levels and never appears in the first three Ladder levels.
- Free Play can include Sudoku or Minesweeper, but avoids selecting both in the same three-puzzle set unless the rest of the set is very easy.

Validation Results

- `node --check` passed for config, symbols, scoring, puzzles, game, lab, and the new validator.
- `scripts/validate-sudoku-minesweeper.js` : 300 samples per new type, Ladder placement, Free Play placement, uniqueness, and targeting passed.
- `scripts/validate-puzzles.js` : 4,500 daily rounds across 300 dates and 39 puzzle types passed.
- `scripts/validate-pattern-systems.js` : 39 total types, 36 active, 300 dates, five attempts passed.
- `scripts/validate-survival.js` , `validate-targeting.js` , `validate-freeplay.js` , `validate-word-puzzles.js` , and `validate-symbol-display.js` passed.
- `git diff --check` passed.

Browser Verification Results

- Served locally with `python3 -m http.server 8080 -d public` and a clean screenshot origin on port `8081` .
- Main app, Free Play, Sudoku lab, and Minesweeper lab loaded with no browser console errors.
- Verified Mini Sudoku Swap selection and correct feedback.
- Verified Free Play wrong Sudoku submission adds a mistake and keeps playing.
- Verified Minesweeper clue cells are disabled, hidden cells are selectable, flags are visible when selected, and correct submissions pause into feedback.
- Verified lab filters for Sudoku and Minesweeper plus answer-hidden/answer-shown states.

Local Run Instructions

```
python3 -m http.server 8080 -d public
# then open http://localhost:8080
# lab: http://localhost:8080/lab.html
```

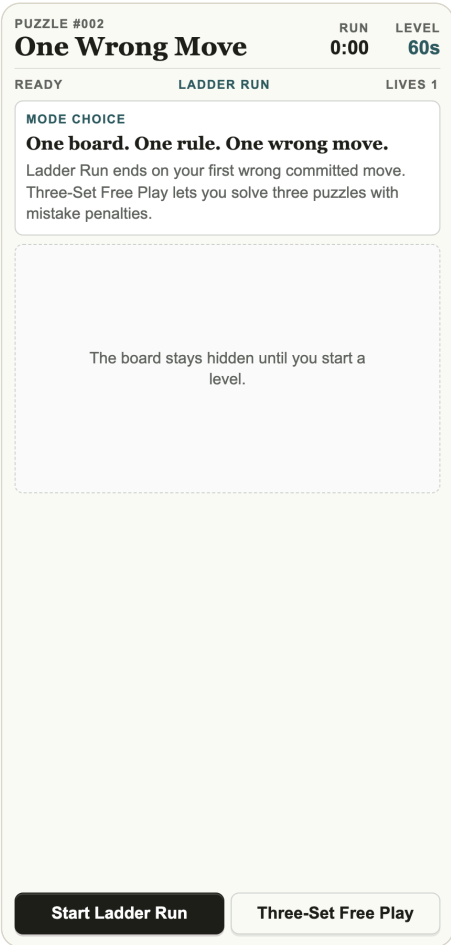
Netlify Deploy Settings

- Build command: blank
- Publish directory: `public`

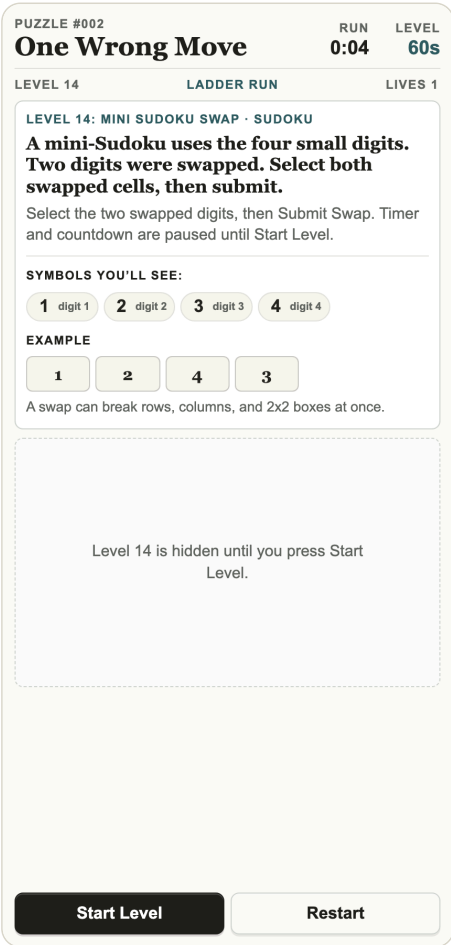
Git Commands Run

```
git switch -c sudoku-minesweeper-puzzles || git switch sudoku-minesweeper-puzzles
node --check public/config.js public/symbols.js public/scoring.js public/puzzles.js public/game.js public/lab.js
node scripts/validate-sudoku-minesweeper.js
node scripts/validate-scoring.js
node scripts/validate-puzzles.js
node scripts/validate-pattern-systems.js
node scripts/validate-survival.js
node scripts/validate-targeting.js
node scripts/validate-freeplay.js
node scripts/validate-word-puzzles.js
node scripts/validate-symbol-display.js
git diff --check
```

Screenshots



Mobile intro / Ladder entry (013-main-intro.png)



Ladder briefing for Mini Sudoku Swap (013-sudoku-swap-briefing.png)

PUZZLE #002

One Wrong Move

RUN0:05

LEVEL60s

LEVEL 14

LADDER RUN

LIVES 1

LEVEL 14: MINI SUDOKU SWAP · SUDOKU

A mini-Sudoku uses the four small digits. Two digits were swapped. Select both swapped cells, then submit.

Select the two swapped digits, then Submit Swap. One wrong move ends the run.

1	2	3	4
4	1	2	1
3	4	1	2
2	3	4	3

Submit Swap

Clear selection

Restart

Mini Sudoku Swap active board (013-sudoku-swap-active.png)

PUZZLE #002

One Wrong Move

RUN0:05

LEVEL60s

LEVEL 14

LADDER RUN

LIVES 1

LEVEL 14: MINI SUDOKU SWAP · SUDOKU

A mini-Sudoku uses the four small digits. Two digits were swapped. Select both swapped cells, then submit.

Select the two swapped digits, then Submit Swap. One wrong move ends the run.

1	2	3	4
4	1.	2	1
3	4	1	2
2	3.	4	3

Submit Swap

Clear selection

Restart

Mini Sudoku Swap selected two-cell swap (013-sudoku-swap-selected.png)

PUZZLE #002

One Wrong Move

RUN0:05

LEVEL60s

LEVEL 14

LADDER RUN

LIVES 1

LEVEL 14: MINI SUDOKU SWAP · SUDOKU

A mini-Sudoku uses the four small digits. Two digits were swapped. Select both swapped cells, then submit.

Select the two swapped digits, then Submit Swap. One wrong move ends the run.

Correct

Swapping those two digits restores every row, column, and 2x2 box.

1	2	3	4
4	1	2	1
3	4	1	2
2	3	4	3

Next Level

Restart

Mini Sudoku Swap correct feedback (013-sudoku-swap-correct-feedback.png)

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 1 OF 3

FREE PLAY · VARIANT 6

MISTAKES 0

PUZZLE 1: SUDOKU CONFLICT · SUDOKU

A mini-Sudoku uses the four small digits. Each row, column, and box needs every digit once. Tap the wrong digit.

Tap one square. Timer and countdown are paused until Start Puzzle.

SYMBOLS YOU'LL SEE:

1 digit 12 digit 23 digit 34 digit 4

EXAMPLE

1234

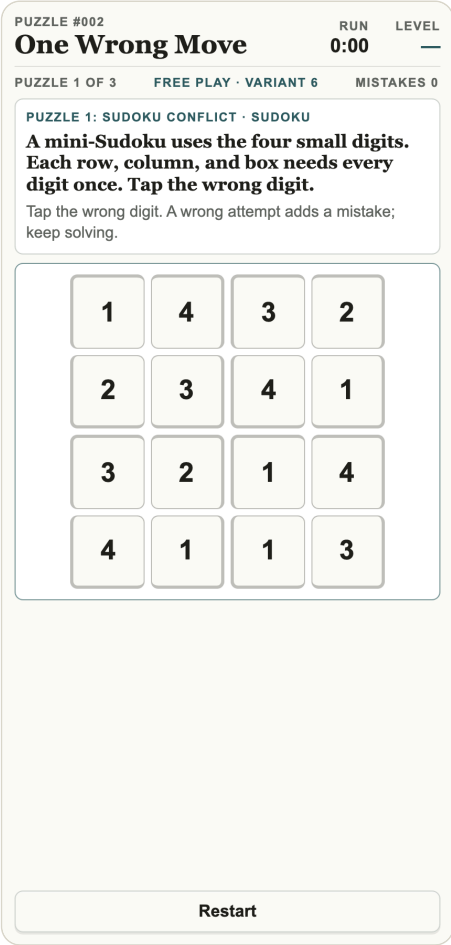
Rows, columns, and 2x2 boxes each need 1, 2, 3, and 4.

Level 1 is hidden until you press Start Level.

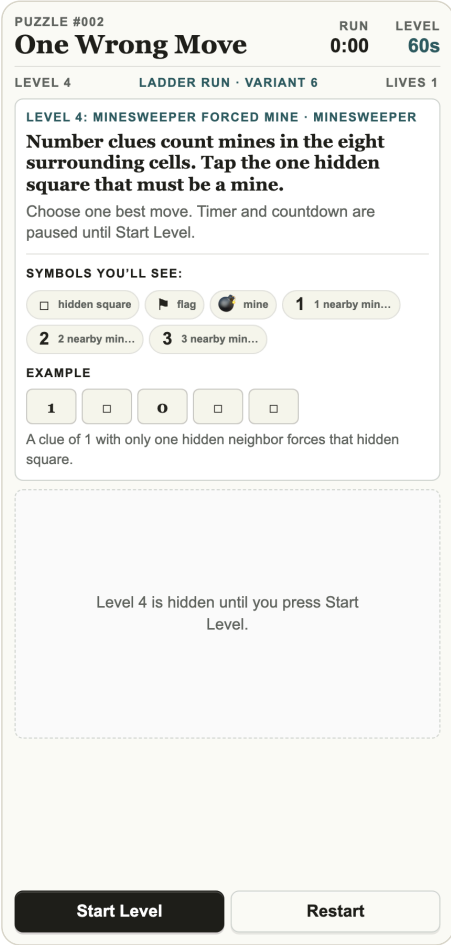
Start Puzzle

Restart

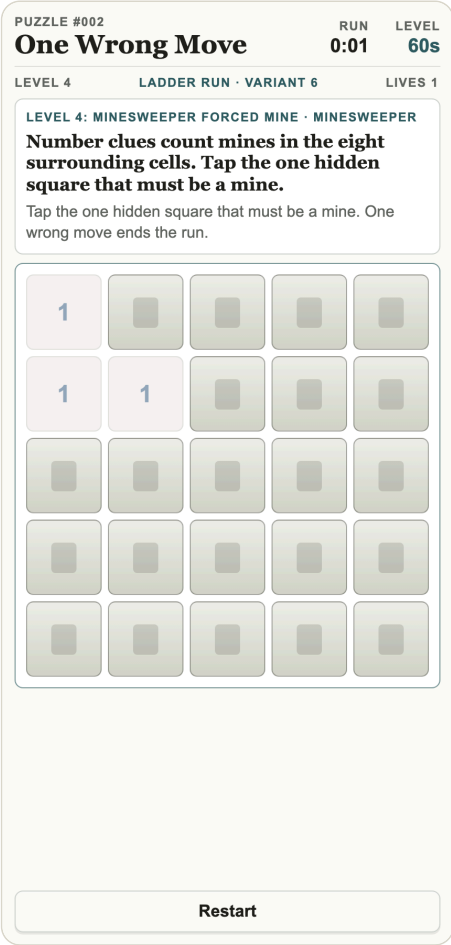
Sudoku Conflict briefing (013-sudoku-conflict-briefing.png)



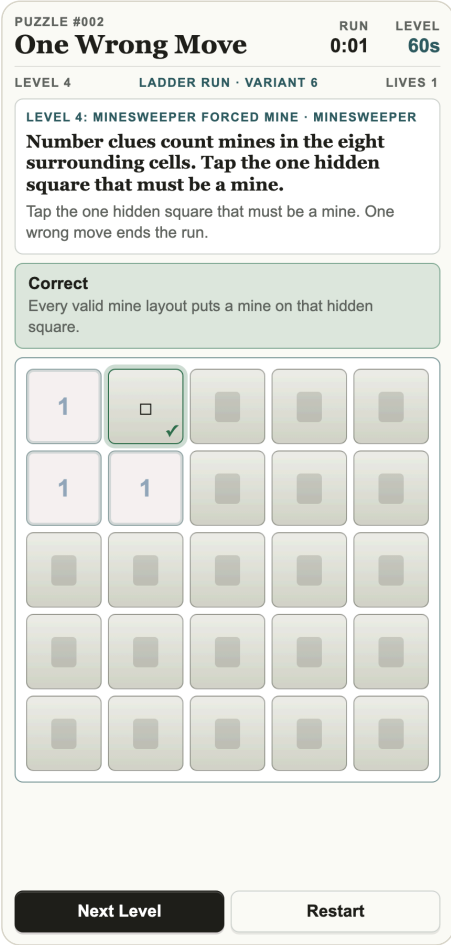
Sudoku Conflict active board (013-sudoku-conflict-active.png)



Minesweeper Forced Mine briefing (013-minesweeper-forced-briefing.png)



Minesweeper Forced Mine active board (013-minesweeper-forced-active.png)



Minesweeper Forced Mine correct feedback (013-minesweeper-forced-correct-feedback.png)

PUZZLE #002

One Wrong Move

RUN0:02

LEVEL60s

LEVEL 9

LADDER RUN · VARIANT 6

LIVES 1

LEVEL 9: MINESWEEPER MARK ALL · MINESWEEPER

Number clues count nearby mines. Select every mine, then submit.

Select every mine, then Submit Flags. Timer and countdown are paused until Start Level.

SYMBOLS YOU'LL SEE:

hidden square

flag

mine

1 1 nearby min...

2 2 nearby min...

3 3 nearby min...

EXAMPLE

1

2

Clues count all eight neighboring cells; flags are your planned mines.

Level 9 is hidden until you press Start Level.

Start Level

Restart

Minesweeper Mark All active board (013-minesweeper-mark-all-active.png)

PUZZLE #002

One Wrong Move

RUN0:03

LEVEL60s

LEVEL 9

LADDER RUN · VARIANT 6

LIVES 1

LEVEL 9: MINESWEEPER MARK ALL · MINESWEEPER

Number clues count nearby mines. Select every mine, then submit.

Select every mine, then Submit Flags. One wrong move ends the run.

0	0	0	0	0
1				1
1				1
1				1
0	0	0	0	0

Submit Flags

Clear flags

Restart

Minesweeper Mark All selected flags (013-minesweeper-flags-selected.png)

file:///tmp/013-sudoku-minesweeper-report.html

8/12

PUZZLE #002

One Wrong Move

RUN0:03

LEVEL60s

LEVEL 9

LADDER RUN · VARIANT 6

LIVES 1

LEVEL 9: MINESWEEPER MARK ALL · MINESWEEPER

Number clues count nearby mines. Select every mine, then submit.

Select every mine, then Submit Flags. One wrong move ends the run.

Correct

Those flags are the only mine set that satisfies every clue.

0	0	0	0	0
1				1
1				1
1				1
0	0	0	0	0

Next Level

Restart

Minesweeper Mark All correct feedback (013-minesweeper-mark-all-correct-feedback.png)

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 2 OF 3

FREE PLAY · VARIANT 2

MISTAKES 0

PUZZLE 2: MINI SUDOKU SWAP · SUDOKU

A mini-Sudoku uses the four small digits. Two digits were swapped. Select both swapped cells, then submit.

Select the two swapped digits, then Submit Swap. Timer and countdown are paused until Start Puzzle.

SYMBOLS YOU'LL SEE:

1 digit 12 digit 23 digit 34 digit 4

EXAMPLE

1243

A swap can break rows, columns, and 2x2 boxes at once.

Level 2 is hidden until you press Start Level.

Start Puzzle

Restart

Free Play using Mini Sudoku Swap (013-freeplay-sudoku-briefing.png)

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 2 OF 3

FREE PLAY · VARIANT 2

MISTAKES 0

PUZZLE 2: MINI SUDOKU SWAP · SUDOKU

A mini-Sudoku uses the four small digits. Two digits were swapped. Select both swapped cells, then submit.

Select the two swapped digits, then Submit Swap. A wrong attempt adds a mistake; keep solving.

2	4	3	1
2	3	4	2
4	2	1	3
3	1	1	4

Submit Swap

Clear selection

Restart

Free Play Sudoku active board (013-freeplay-sudoku-active.png)

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 2 OF 3

FREE PLAY · VARIANT 2

MISTAKES 1

PUZZLE 2: MINI SUDOKU SWAP · SUDOKU

A mini-Sudoku uses the four small digits. Two digits were swapped. Select both swapped cells, then submit.

Select the two swapped digits, then Submit Swap. A wrong attempt adds a mistake; keep solving.

Try again

Choose exactly two cells: the pair that should trade digits to make all rows, columns, and boxes valid.

2	4	3	1
2	3	4	2
4	2	1	3
3	1	1	4

Submit Swap

Clear selection

Restart

Free Play wrong Sudoku submission keeps playing (013-freeplay-wrong-sudoku-keeps-playing.png)

PUZZLE #002

One Wrong Move

RUN0:01

LEVEL—

COMPLETE

FREE PLAY · VARIANT 2

MISTAKES 1

THREE-SET FREE PLAY COMPLETE

Solved 3 of 3

Lower is better: rounded-up solving time plus 10 seconds per mistake.

S	—	—	L ✓	
			L	7
				F

LEVELS

3 of 3

RUN TIME

0:01

ENDED ON

2s

RESULT

12s

MORE LEVELS IS BETTER. TIES GO TO FASTER RUNS.

End reason

1

Last puzzle

1 × 10s = +10s

Limit

Lower is better

Free Play lower-is-better result screen (013-freeplay-results.png)

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

LEVEL LIMIT

60

FILTER

Sudoku

Regenerate

Ladder stream preview

First 20 generated levels for this seed. Limit: 60s/level.

1. Object Rack Complete · Objects · twoStep2. Pair Pact · Relationships · identifyOne3. Compass Rose · Compass · identifyOne4. Card Straight · Cards · identifyOne5. Train Route · Routes · identifyOne6. Recipe Swap · Food · multiSelect7. Chess Attack · Chess · identifyOne8. Crossword Pair · Crossword · multiSelect9. Mini Sudoku Swap · Sudoku · multiSelect10. Go Liberties · Go · multiSelect11. Yahtzee Fix · Yahtzee · multiSelect12. Chess Attack · Chess · identifyOne13. Minesweeper Forced Mine · Minesweeper · chooseOne14. Maze Bridge Repair · Maze · multiSelect15. Circuit Switch Pair · Circuits · multiSelect16. Tetris Fit · Tetris · multiSelect17. Mini Sudoku Swap · Sudoku · multiSelect18. Checkers Jump · Checkers · identifyOne19. Crossword Pair · Crossword · multiSelect20. Maze Key Exit · Maze · twoStep

Three-Set Free Play preview

Three puzzles. Wrong attempts add mistakes. Score = cell(active seconds) + mistakes × 10s.

1. Sudoku Conflict · Sudoku · identifyOne · cell2. Pair Pact · Relationships · identifyOne · cell3. Compass Rose · Compass · identifyOne · cell4. Card Straight · Cards · identifyOne · cell5. Train Route · Routes · identifyOne · cell6. Recipe Swap · Food · multiSelect · cell7. Chess Attack · Chess · identifyOne · cell8. Crossword Pair · Crossword · multiSelect · cell9. Mini Sudoku Swap · Sudoku · multiSelect · cell10. Go Liberties · Go · multiSelect · cell11. Yahtzee Fix · Yahtzee · multiSelect · cell12. Chess Attack · Chess · identifyOne · cell13. Minesweeper Forced Mine · Minesweeper · chooseOne · cell14. Maze Bridge Repair · Maze · multiSelect · cell15. Circuit Switch Pair · Circuits · multiSelect · cell16. Tetris Fit · Tetris · multiSelect · cell17. Mini Sudoku Swap · Sudoku · multiSelect · cell18. Checkers Jump · Checkers · identifyOne · cell19. Crossword Pair · Crossword · multiSelect · cell20. Maze Key Exit · Maze · twoStep

Puzzle lab Sudoku filter / section (013-lab-sudoku.png)

DEVELOPMENT LAB

Game

One Wrong Move Lab

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

LEVEL LIMIT

60

FILTER

Minesweeper

Regenerate

Ladder stream preview

First 20 generated levels for this seed. Limit: 60s/level.

1. Object Rack Complete · Objects · twoStep2. Pair Pact · Relationships · identifyOne3. Compass Rose · Compass · identifyOne4. Card Straight · Cards · identifyOne5. Train Route · Routes · identifyOne6. Recipe Swap · Food · multiSelect7. Chess Attack · Chess · identifyOne8. Crossword Pair · Crossword · multiSelect9. Mini Sudoku Swap · Sudoku · multiSelect10. Go Liberties · Go · multiSelect11. Yahtzee Fix · Yahtzee · multiSelect12. Chess Attack · Chess · identifyOne13. Minesweeper Forced Mine · Minesweeper · chooseOne14. Maze Bridge Repair · Maze · multiSelect15. Circuit Switch Pair · Circuits · multiSelect16. Tetris Fit · Tetris · multiSelect17. Mini Sudoku Swap · Sudoku · multiSelect18. Checkers Jump · Checkers · identifyOne19. Crossword Pair · Crossword · multiSelect20. Maze Key Exit · Maze · twoStep

Three-Set Free Play preview

Three puzzles. Wrong attempts add mistakes. Score = ceil(active seconds) + mistakes × 10s.

1. Sudoku Conflict · Sudoku · identifyOne · cell2.

Puzzle lab Minesweeper answer hidden (013-lab-minesweeper-hidden.png)

DEVELOPMENT LAB

Game

One Wrong Move Lab

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

LEVEL LIMIT

60

FILTER

All puzzles

Regenerate

Ladder stream preview

First 20 generated levels for this seed. Limit: 60s/level.

1. Object Rack Complete · Objects · twoStep2. Pair Pact · Relationships · identifyOne3. Compass Rose · Compass · identifyOne4. Card Straight · Cards · identifyOne5. Train Route · Routes · identifyOne6. Recipe Swap · Food · multiSelect7. Chess Attack · Chess · identifyOne8. Crossword Pair · Crossword · multiSelect9. Mini Sudoku Swap · Sudoku · multiSelect10. Go Liberties · Go · multiSelect11. Yahtzee Fix · Yahtzee · multiSelect12. Chess Attack · Chess · identifyOne13. Minesweeper Forced Mine · Minesweeper · chooseOne14. Maze Bridge Repair · Maze · multiSelect15. Circuit Switch Pair · Circuits · multiSelect16. Tetris Fit · Tetris · multiSelect17. Mini Sudoku Swap · Sudoku · multiSelect18. Checkers Jump · Checkers · identifyOne19. Crossword Pair · Crossword · multiSelect20. Maze Key Exit · Maze · twoStep

Three-Set Free Play preview

Three puzzles. Wrong attempts add mistakes. Score = ceil(active seconds) + mistakes × 10s.

1. Sudoku Conflict · Sudoku · identifyOne · cell2.

Puzzle lab Minesweeper answer shown (013-lab-minesweeper-shown.png)